OSITO DRONE STEM ACADEMY

SERIES 07: BILINGUAL ENGLISH & ESPAÑOL DRONE STEM - AGES 4-9 (BILINGUAL K-3)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 302: Mateo - Colores del Cielo - Sky Colors (Bilingue)

Featured STEM Pilot: Mateo (Junior Aviator & Community Garden Explorer)

Flight Environment: Harbor Sailboat & Bridge Bay - Target Age: Ages 4-9 (Bilingual K-3)

Core STEM & Math Focus: Color Matching Bilingual Vocabulary: Rojo, Azul, Oro

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

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OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

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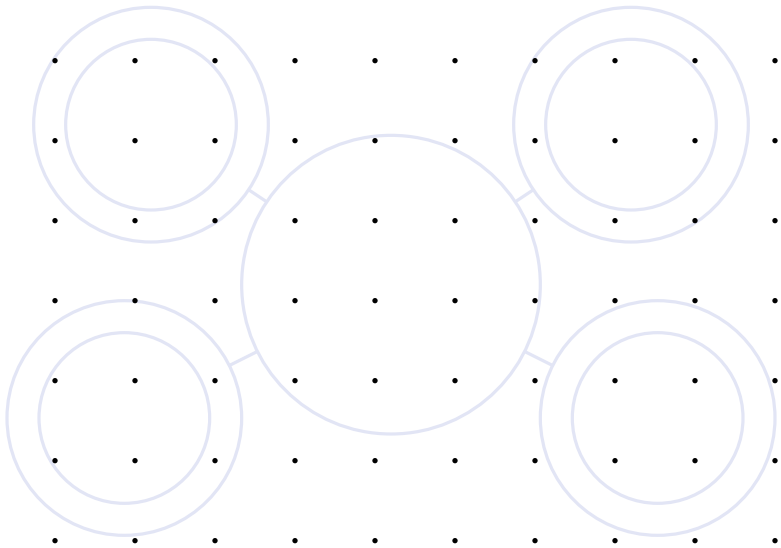
Designed to comply with FAA TRUST youth aviation educational benchmarks.

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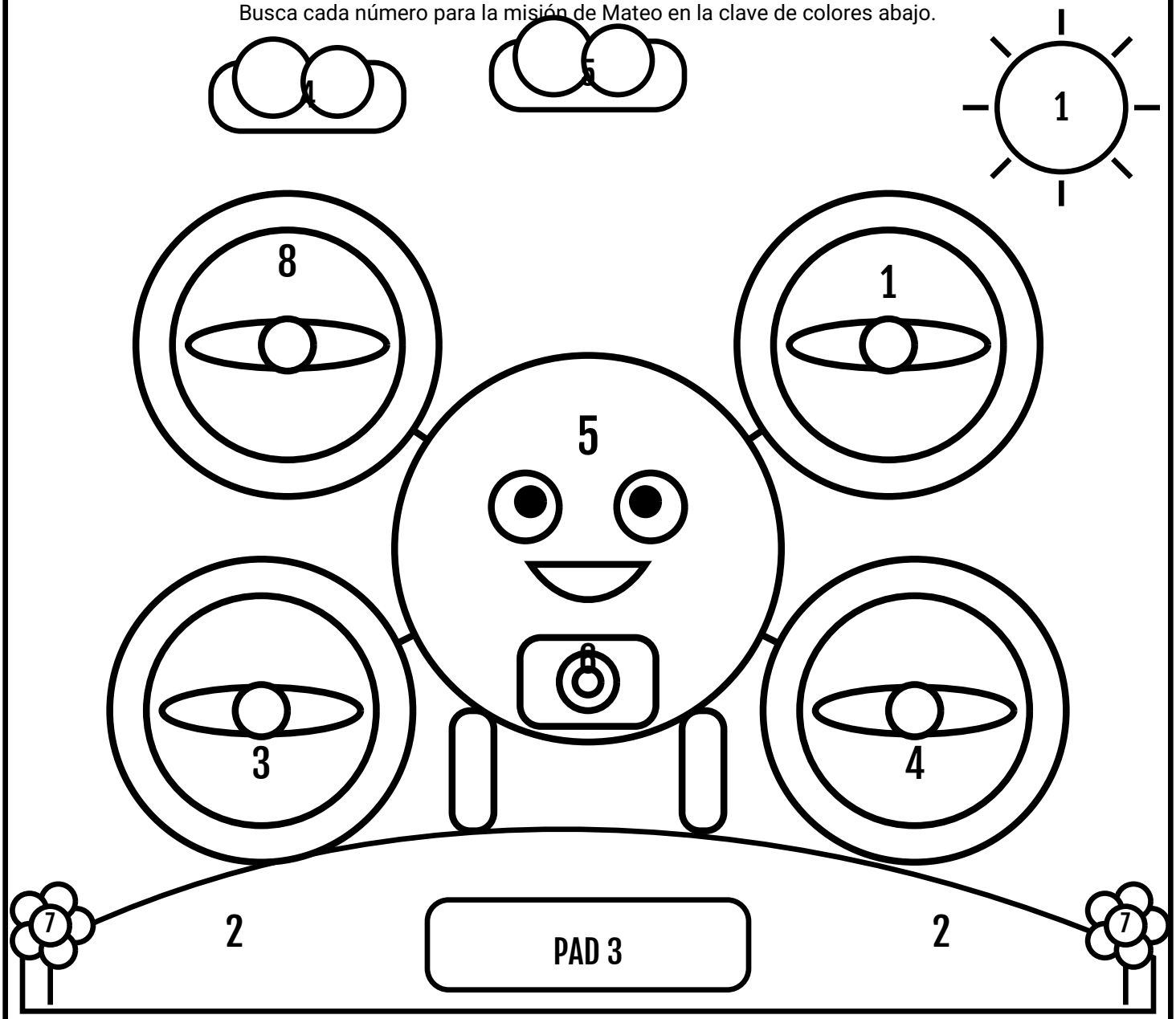


MATEO & PET DRONE – COLORES DEL CIELO – SKY COLORS



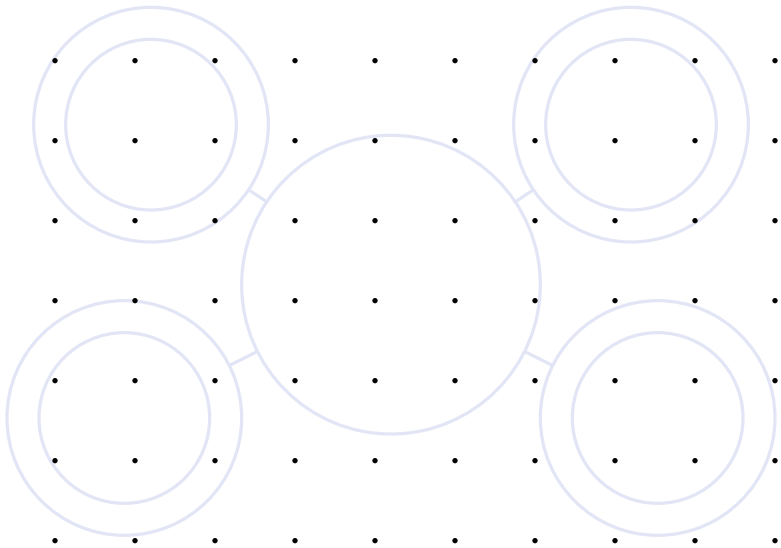
COLOREA POR NÚMEROS – COLOR BY NUMBER DRONE

Busca cada número para la misión de Mateo en la clave de colores abajo.



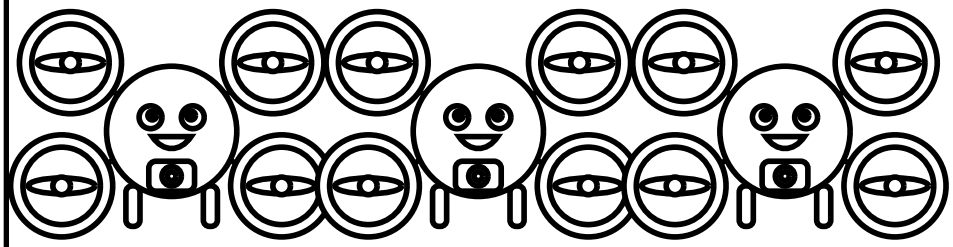
CLAVE DE COLORES / COLOR KEY

1 = ROJO / RED	2 = AZUL / BLUE	3 = AMARILLO / YELLOW	4 = VERDE / GREEN
5 = NARANJA / ORANGE	6 = MORADO / PURPLE	7 = CELESTE / TEAL	8 = CAFÉ / BROWN



MATEMÁTICAS: MISIÓN DE MATEO

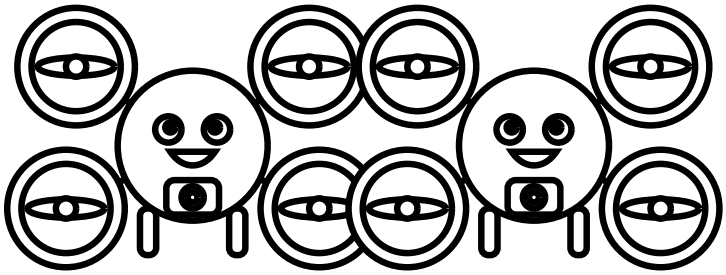
Every quadcopter has 4 ducted propellers! Solve Mateo's flight problems and trace the numbers.



MISSION #1: ALTITUDE MATH

Climb 6ft + 4ft Higher

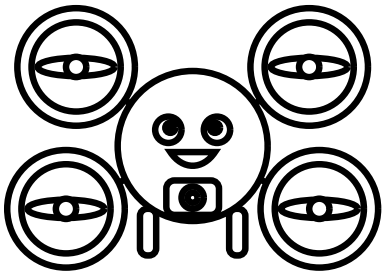
Color the drones & trace numbers!



MISSION #2: ALTITUDE MATH

Climb 4ft + 5ft Higher

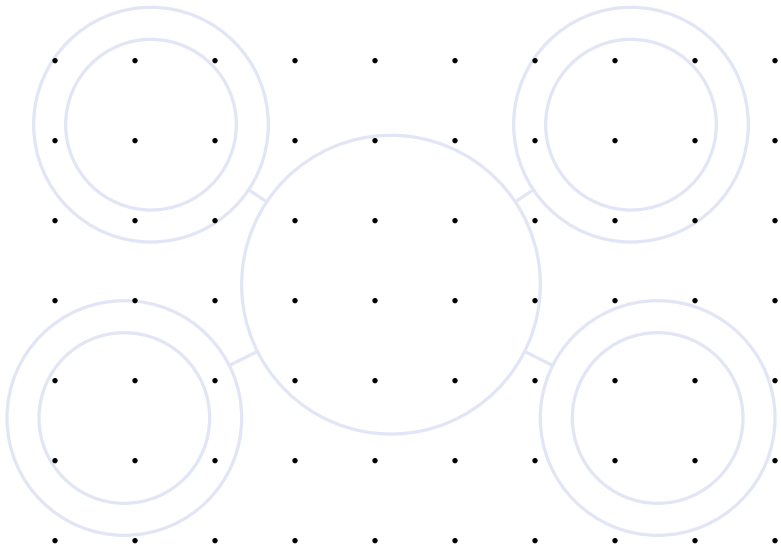
Color the drones & trace numbers!



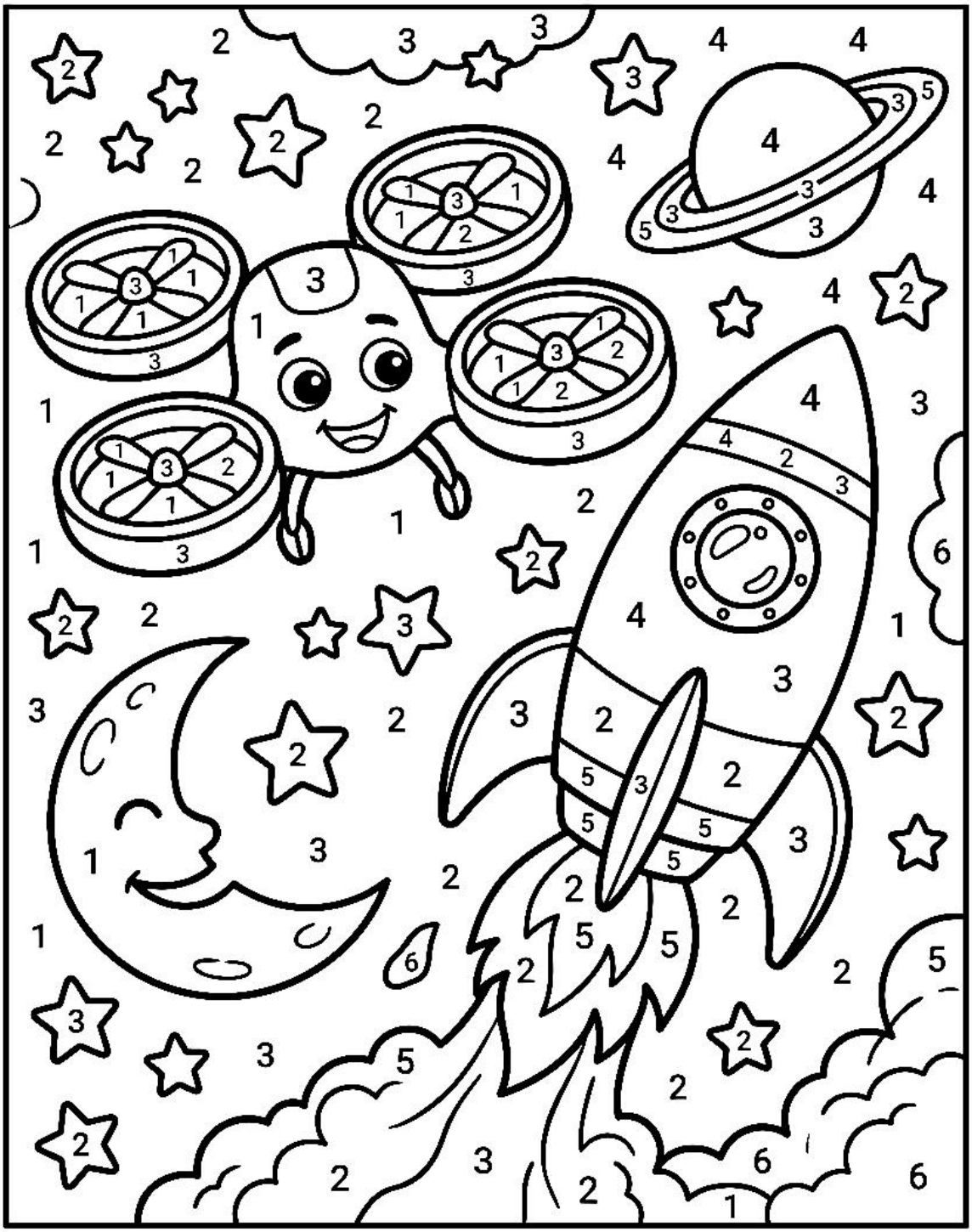
MISSION #3: ALTITUDE MATH

Climb 2ft + 6ft Higher

Color the drones & trace numbers!

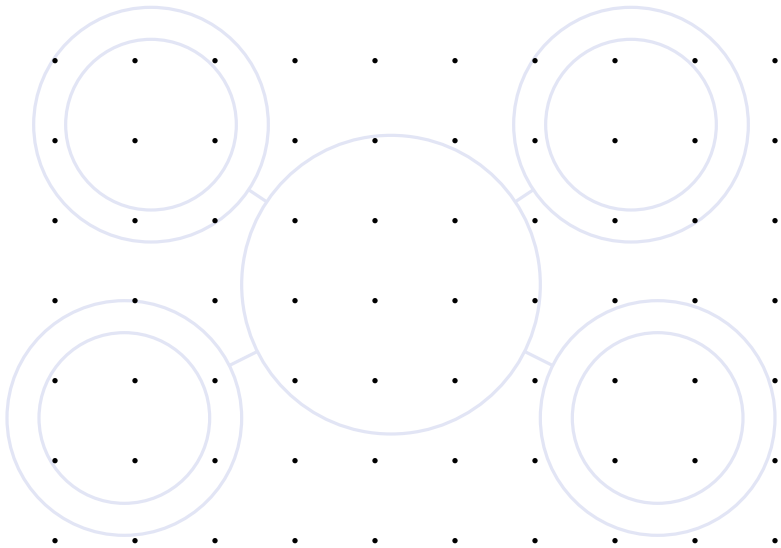


COLOR BY NUMBER: SPACE ADVENTURE



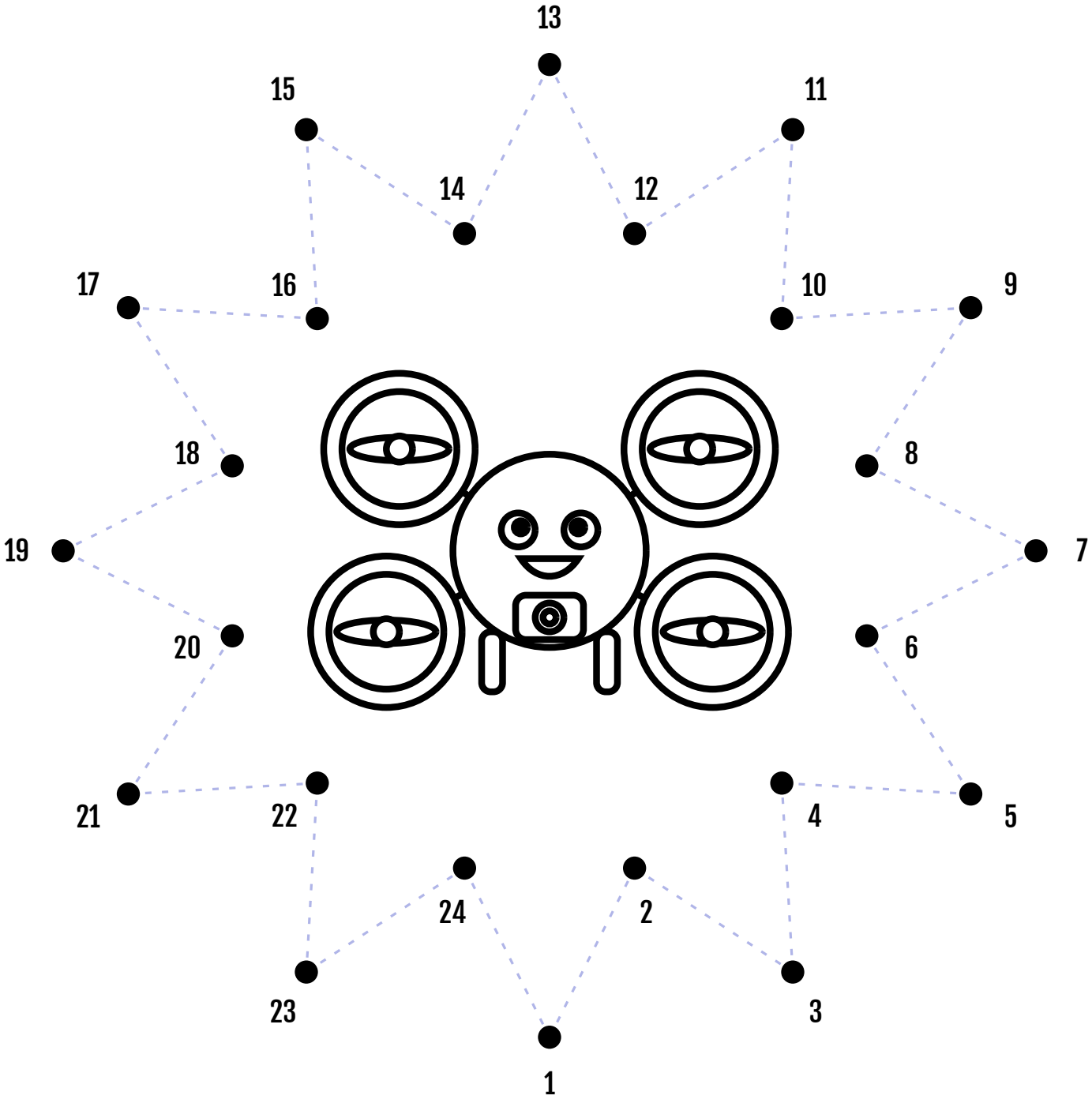
1=Blue 2=Yellow 3=Red 4=Orange 5=Green 6=Black

COLOR BY NUMBER & COUNT ROTORS WITH MATEO!

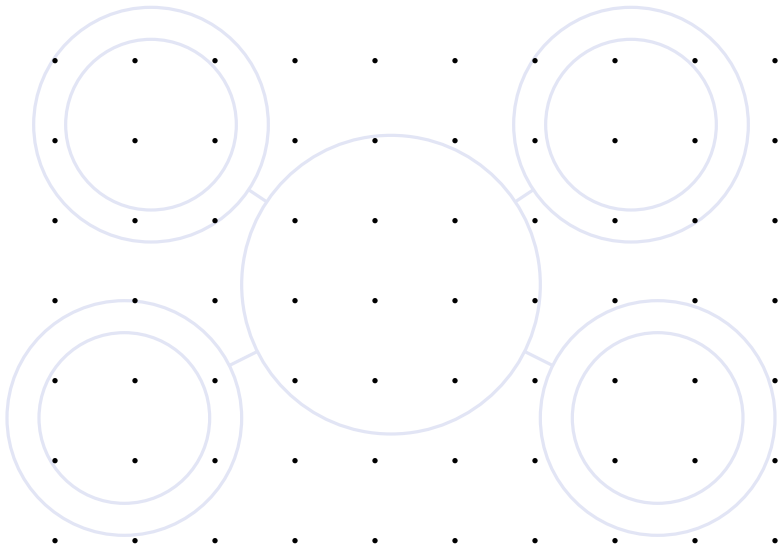


CONNECT THE DOTS: MATEO'S COLORES DEL CIELO – SKY COLORS

Connect the numbered dots in order counting by 1s to complete the aerospace blueprint!



BONUS FOR MATEO: You connected 24 dots counting by 1s! Final number = 24!



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

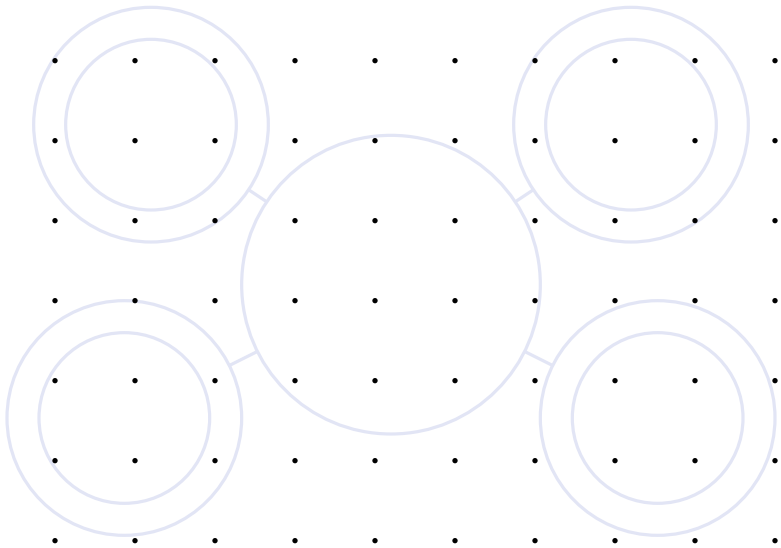
Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

	1	2	3	4	5	6	7
A	4	2	2	2	2	2	4
B	2	5	1	5	1	5	2
C	2	5	1	5	1	5	2
D	2	5	1	1	1	5	2
E	2	5	1	5	1	5	2
F	2	5	1	5	1	5	2
G	4	2	2	2	2	2	4

PIXEL CODE COLOR KEY

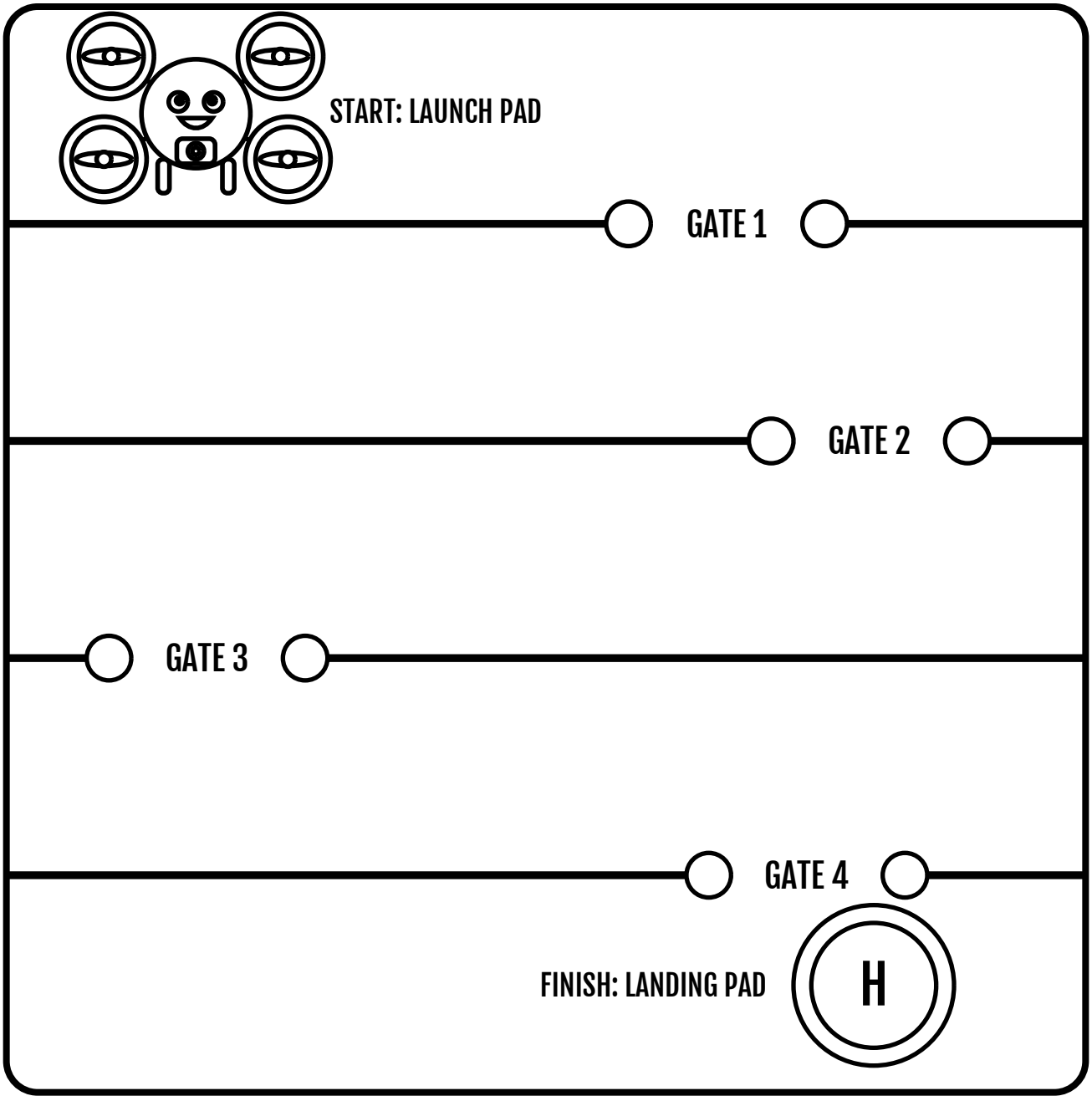
1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!

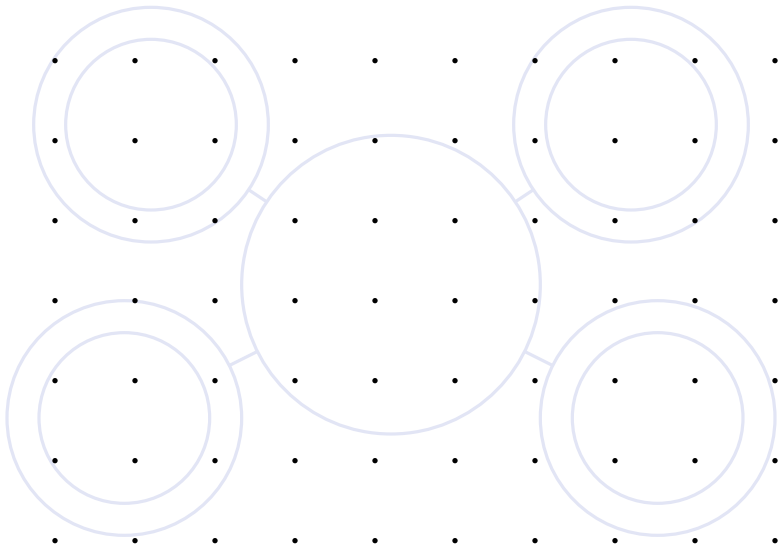


SAFE FLIGHT-PATH MAZE: GUIDE MATEO'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!

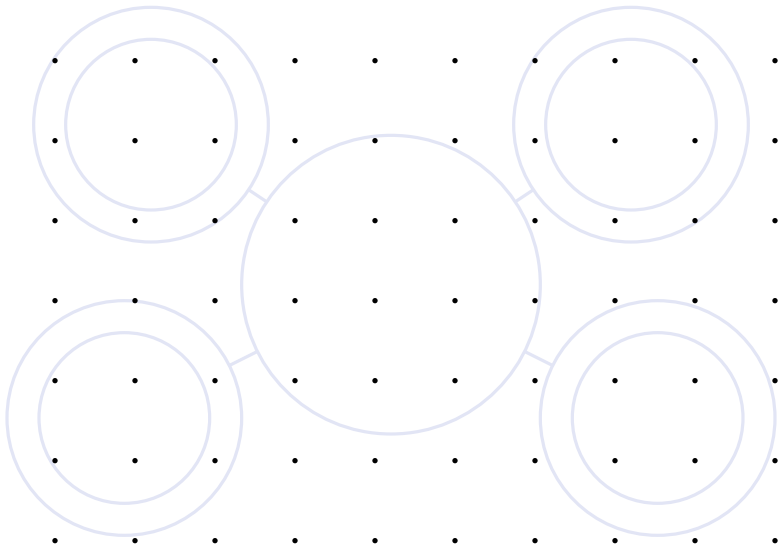


COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!



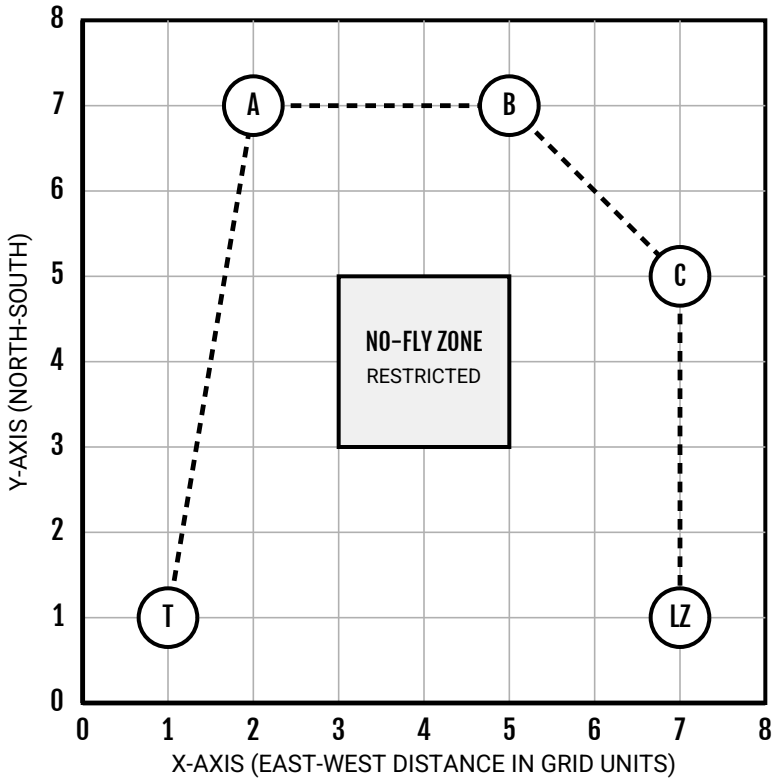


SAFE DRONE FLIGHT AT HARBOR SAILBOAT & BRIDGE BAY



COORDINATE FLIGHT PLAN: MATEO'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Mateo's Colores del Cielo - Sky Colors!

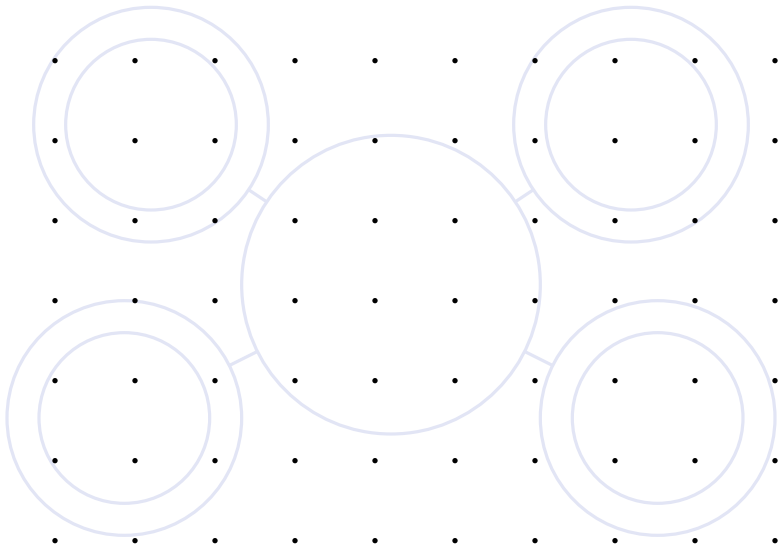


FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Mateo Survey	(2 , 7)
WP B:	Waypoint Colores del	(5 , 7)
WP C:	Bridge Check	(7 , 5)
WP LZ:	Landing Zone	(7 , 1)

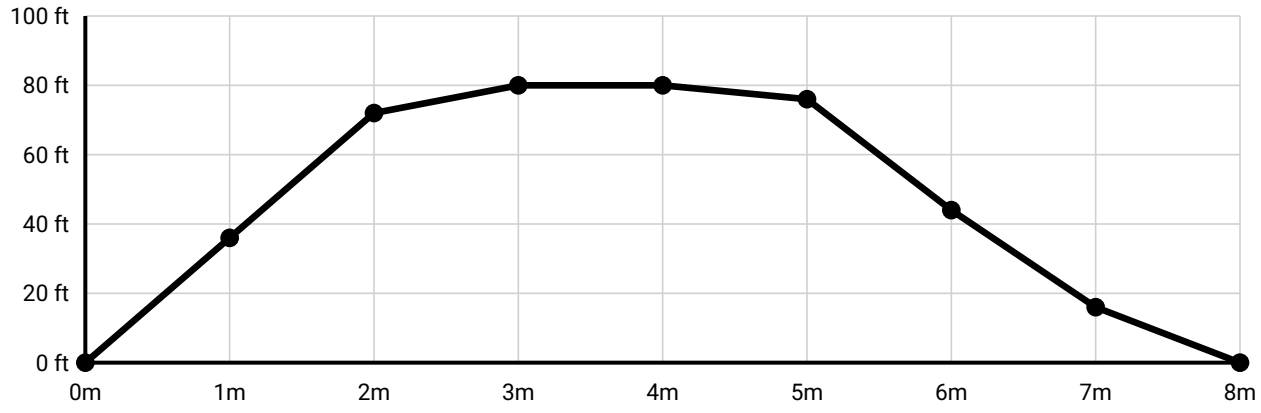
COORDINATE CHALLENGE:

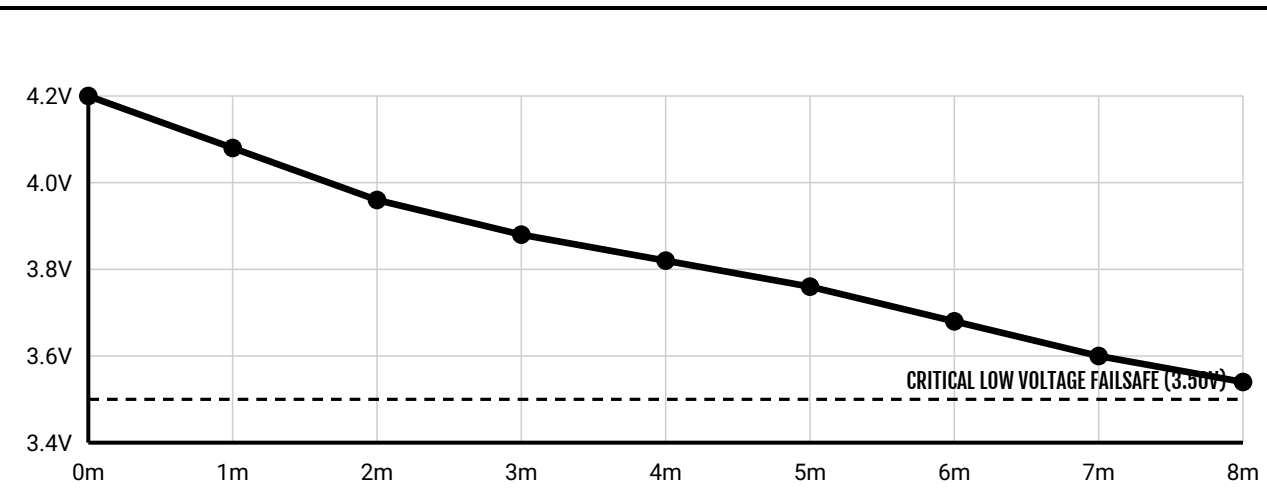
1. Waypoint B coordinates: (5 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

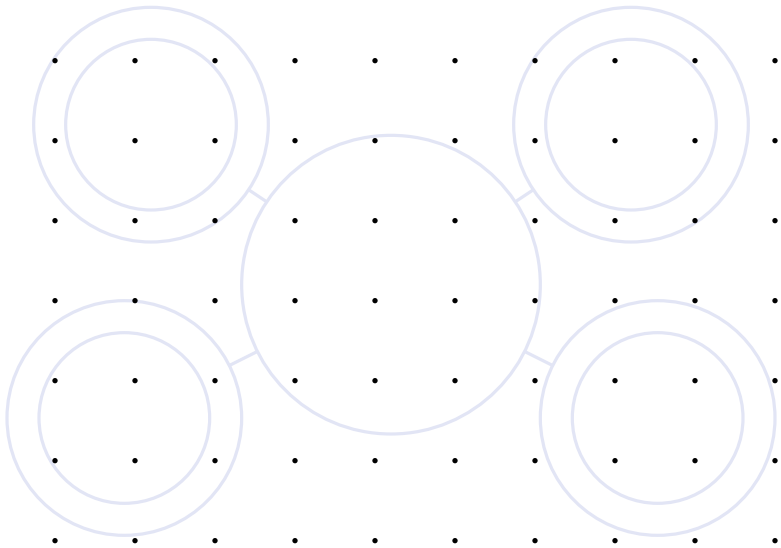


FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.

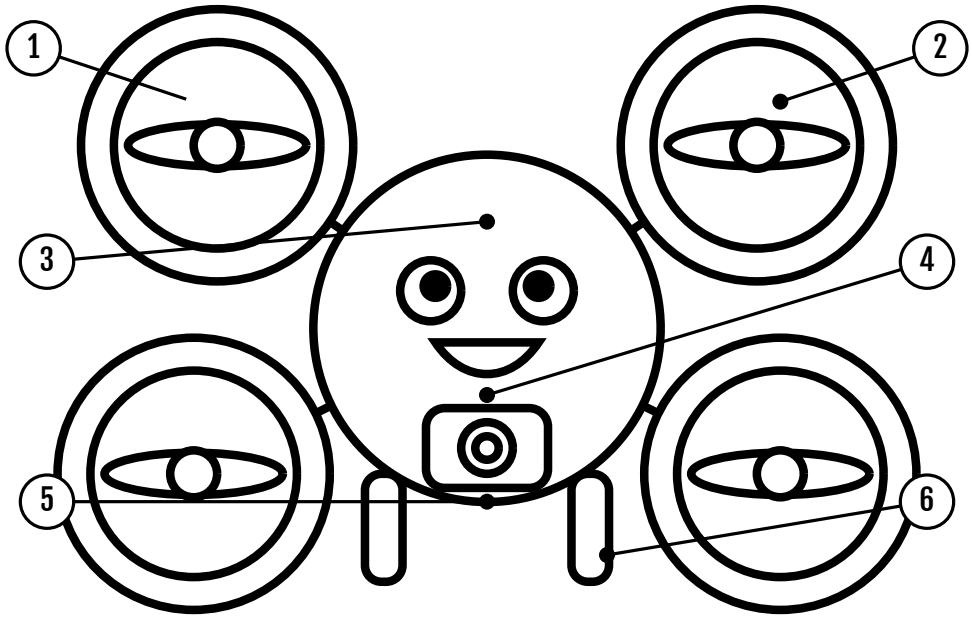






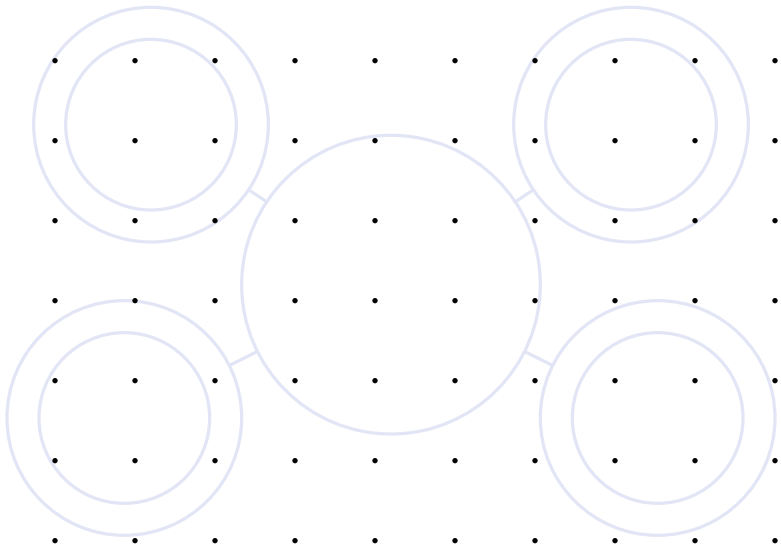
AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



1	**Ducted Prop Guards**	Duct rings that shield propellers and provide 360-degree impact safety.
2	**Brushless DC Motors**	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	**Flight Controller (FC)**	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	**4-in-1 ESC Board**	Electronic Speed Controller regulating power sent from battery to each motor.
5	**Micro FPV Camera**	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	**1S LiPo Battery**	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

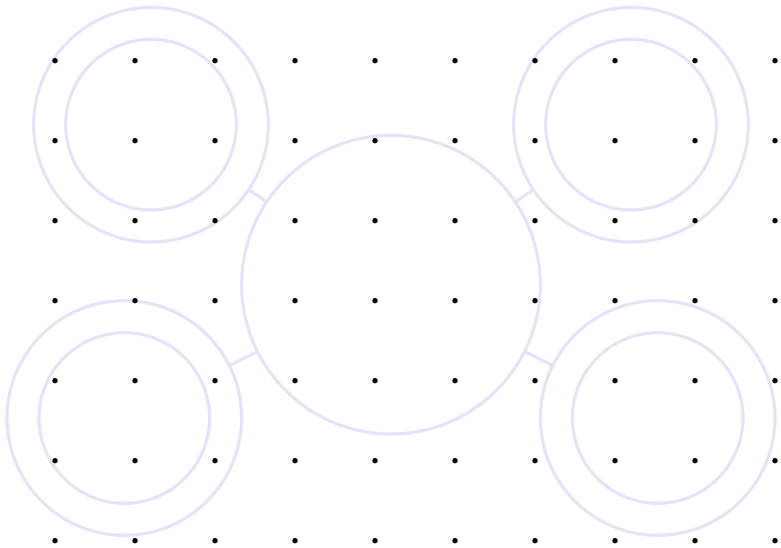
CUB-75 SPECS: Wheelbase: 75mm – AUW Weight: 39.5g – Motors: 0802 19000KV – Flight Time: 8.0 min



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.

BLEED-PREVENTION BACKING PAGE 26 - BOOK #302 - FLIGHT LOG & SKETCH PAD



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #302: COLORES DEL CIELO – SKY COLORS

Core Aeronautical Skill: Color Matching Bilingual Vocabulary: Rojo, Azul, Oro

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

